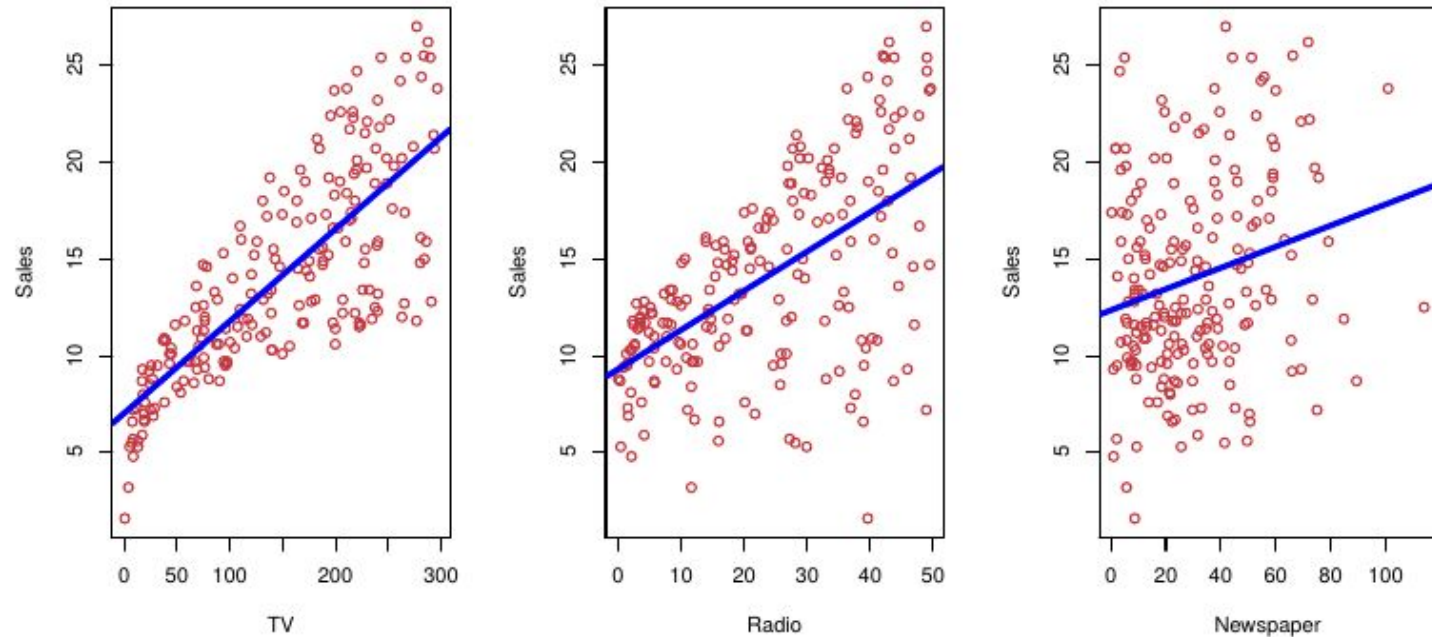


What is Statistical Learning?



Shown are **Sales** vs **TV**, **Radio** and **Newspaper**, with a blue linear-regression line fit separately to each.

Can we predict **Sales** using these three?

Perhaps we can do better using a model

$$\text{Sales} \approx f(\text{TV}, \text{Radio}, \text{Newspaper})$$

TV	radio	newspaper	sales
230.1	37.8	69.2	22.1
44.5	39.3	45.1	10.4
17.2	45.9	69.3	9.3
151.5	41.3	58.5	18.5

Notation

Here **Sales** is a *response* or *target* that we wish to predict. We generically refer to the response as Y .

TV is a *feature*, or *input*, or *predictor*; we name it X_1 .

Likewise name **Radio** as X_2 , and so on.

We can refer to the *input vector* collectively as

$$X = \begin{pmatrix} X_1 \\ X_2 \\ X_3 \end{pmatrix}$$

Now we write our model as

$$Y = f(X) + \epsilon$$

where ϵ captures measurement errors and other discrepancies.

	TV	radio	newspaper		sales
f	(230.1	37.8	69.2	$\sim Y_1$	22.1
f	(44.5	39.3	45.1	$\sim Y_2$	10.4
f	(17.2	45.9	69.3	$\sim Y_3$	9.3
f	(151.5	41.3	58.5	$\sim Y_4$	18.5

	TV	radio	newspaper	
f	(230.1	37.8	69.2	$\} + \epsilon_1 = Y_1$
f	(44.5	39.3	45.1	$\} + \epsilon_2 = \cdot$
f	(17.2	45.9	69.3	$\} + \epsilon_3 = \cdot$
f	(151.5	41.3	58.5	$\} + \epsilon_4 = \cdot$

$$Y = f(X) + \epsilon$$

Notation Continued

In general, we will let x_{ij} represent the value of the j th variable for the i th observation, where $i = 1, 2, \dots, n$ and $j = 1, 2, \dots, p$. Throughout this book, i will be used to index the samples or observations (from 1 to n) and j will be used to index the variables (from 1 to p). We let $\mathbf{X}$ denote an $n \times p$ matrix whose (i, j) th element is x_{ij} . That is,

$$\mathbf{X} = \begin{pmatrix} x_{11} & x_{12} & \dots & x_{1p} \\ x_{21} & x_{22} & \dots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \dots & x_{np} \end{pmatrix}.$$

At times we will be interested in the rows of $\mathbf{X}$, which we write as $x_1, x_2, \dots, x_n$. Here x_i is a vector of length p , containing the p variable measurements for the i th observation. That is,

$$x_i = \begin{pmatrix} x_{i1} \\ x_{i2} \\ \vdots \\ x_{ip} \end{pmatrix}. \tag{1.1}$$

At other times we will instead be interested in the columns of $\mathbf{X}$, which we write as $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_p$. Each is a vector of length n . That is,

$$\mathbf{x}_j = \begin{pmatrix} x_{1j} \\ x_{2j} \\ \vdots \\ x_{nj} \end{pmatrix}.$$

For example, for the **Wage** data, $\mathbf{x}_1$ contains the $n = 3,000$ values for **year**.

Using this notation, the matrix $\mathbf{X}$ can be written as

$$\mathbf{X} = (\mathbf{x}_1 \quad \mathbf{x}_2 \quad \cdots \quad \mathbf{x}_p),$$

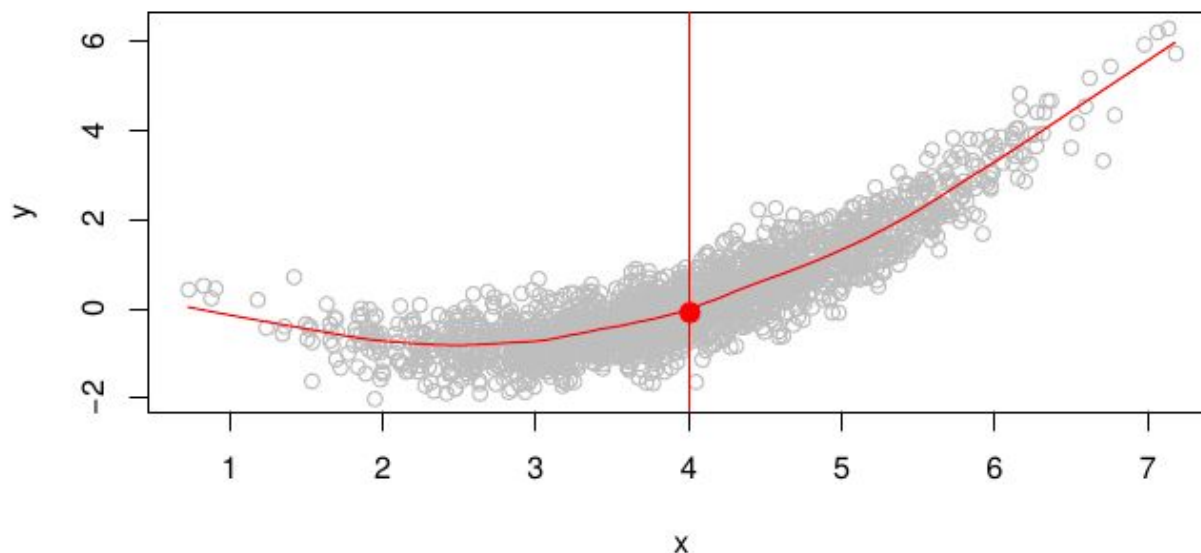
or

$$\mathbf{X} = \begin{pmatrix} x_1^T \\ x_2^T \\ \vdots \\ x_n^T \end{pmatrix}.$$

What is $f(X)$ good for?

(back to sales prediction)

- With a good f we can make predictions of Y at new points $X = x$.
- We can understand which components of $X = (X_1, X_2, \dots, X_p)$ are important in explaining Y , and which are irrelevant. e.g. **Seniority** and **Years of Education** have a big impact on **Income**, but **Marital Status** typically does not.
- Depending on the complexity of f , we may be able to understand how each component X_j of X affects Y .



Is there an ideal $f(X)$? In particular, what is a good value for $f(X)$ at any selected value of X , say $X = 4$? There can be many Y values at $X = 4$. A good value is

$$f(4) = E(Y|X = 4)$$

$E(Y|X = 4)$ means *expected value* (average) of Y given $X = 4$.

This ideal $f(x) = E(Y|X = x)$ is called the *regression function*.

The regression function $f(x)$

- Is also defined for vector X ; e.g.

$$f(x) = f(x_1, x_2, x_3) = E(Y|X_1 = x_1, X_2 = x_2, X_3 = x_3)$$

The regression function $f(x)$

- Is also defined for vector X ; e.g.
$$f(x) = f(x_1, x_2, x_3) = E(Y|X_1 = x_1, X_2 = x_2, X_3 = x_3)$$
- Is the *ideal* or *optimal* predictor of Y with regard to mean-squared prediction error: $f(x) = E(Y|X = x)$ is the function that minimizes $E[(Y - g(X))^2|X = x]$ over all functions g at all points $X = x$.

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- $\epsilon = Y - f(x)$ is the *irreducible* error — i.e. even if we knew $f(x)$, we would still make errors in prediction, since at each $X = x$ there is typically a distribution of possible Y values.

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- $\epsilon = Y - f(x)$ is the *irreducible* error — i.e. even if we knew $f(x)$, we would still make errors in prediction, since at each $X = x$ there is typically a distribution of possible Y values.
- For any estimate $\hat{f}(x)$ of $f(x)$, we have

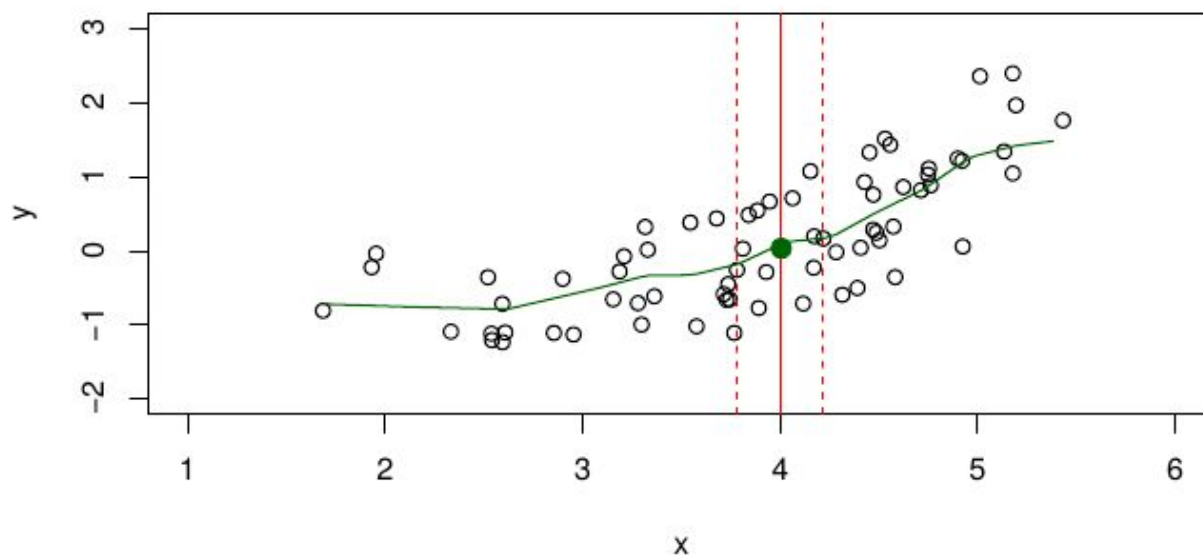
$$E[(Y - \hat{f}(X))^2|X = x] = \underbrace{[f(x) - \hat{f}(x)]^2}_{\text{Reducible}} + \underbrace{\text{Var}(\epsilon)}_{\text{Irreducible}}$$

How to estimate f

- Typically we have few if any data points with $X = 4$ exactly.
- So we cannot compute $E(Y|X = x)$!
- Relax the definition and let

$$\hat{f}(x) = \text{Ave}(Y|X \in \mathcal{N}(x))$$

where $\mathcal{N}(x)$ is some *neighborhood* of x .



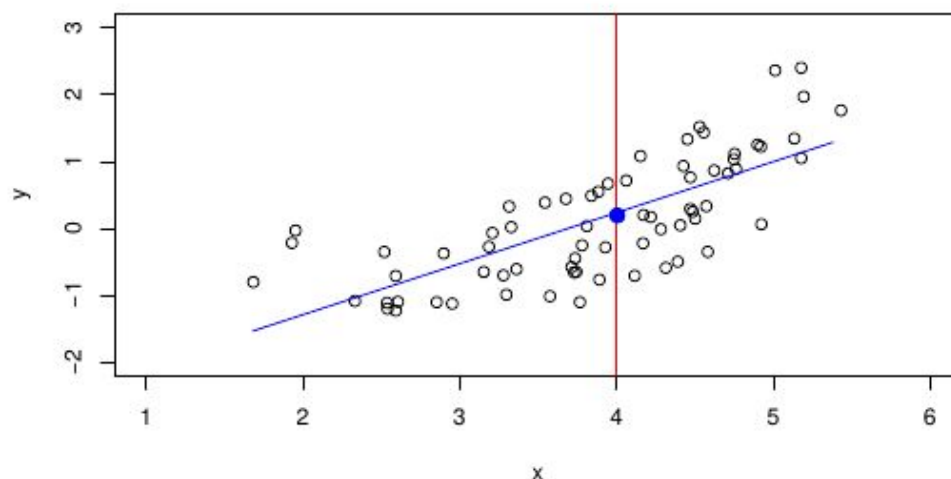
Parametric and structured models

The *linear* model is an important example of a parametric model:

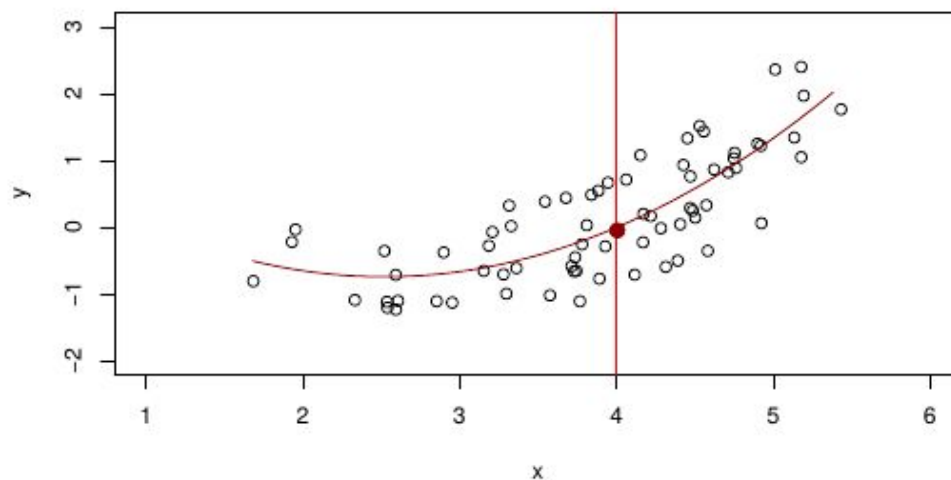
$$f_L(X) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots \beta_p X_p.$$

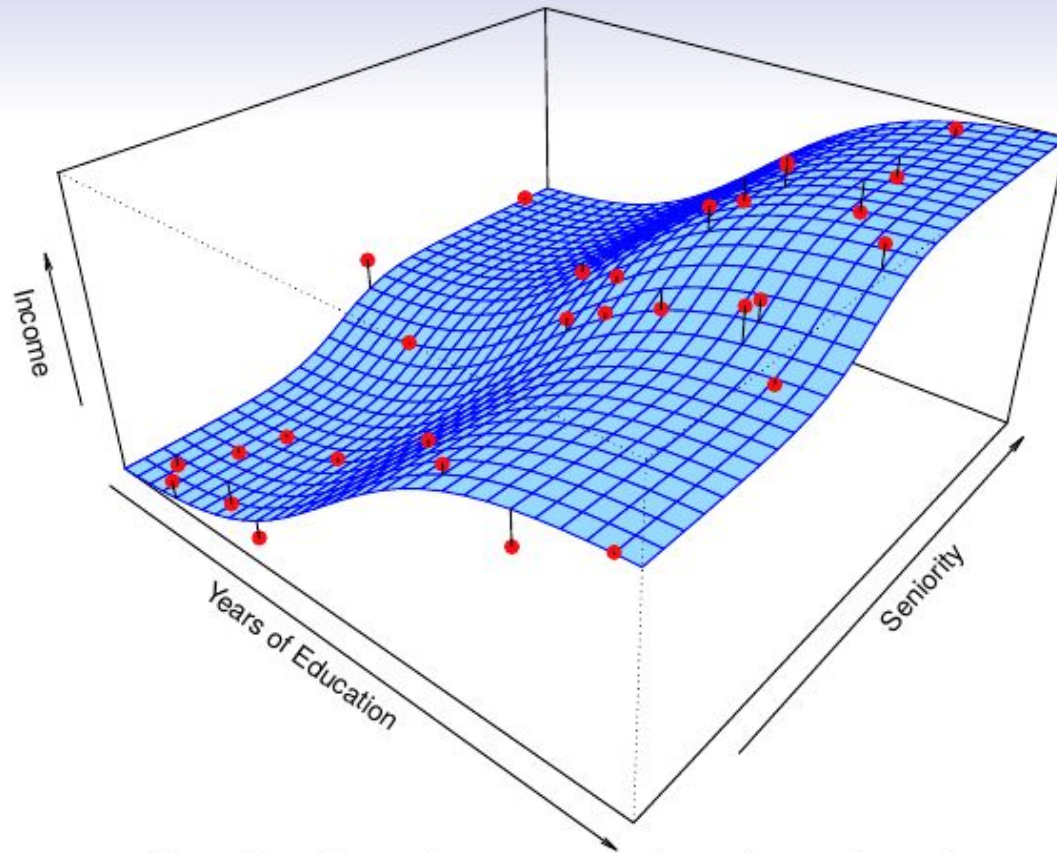
- A linear model is specified in terms of $p + 1$ parameters $\beta_0, \beta_1, \dots, \beta_p$.
- We estimate the parameters by fitting the model to training data.
- Although it is *almost never correct*, a linear model often serves as a good and interpretable approximation to the unknown true function $f(X)$.

A linear model $\hat{f}_L(X) = \hat{\beta}_0 + \hat{\beta}_1 X$ gives a reasonable fit here



A quadratic model $\hat{f}_Q(X) = \hat{\beta}_0 + \hat{\beta}_1 X + \hat{\beta}_2 X^2$ fits slightly better.

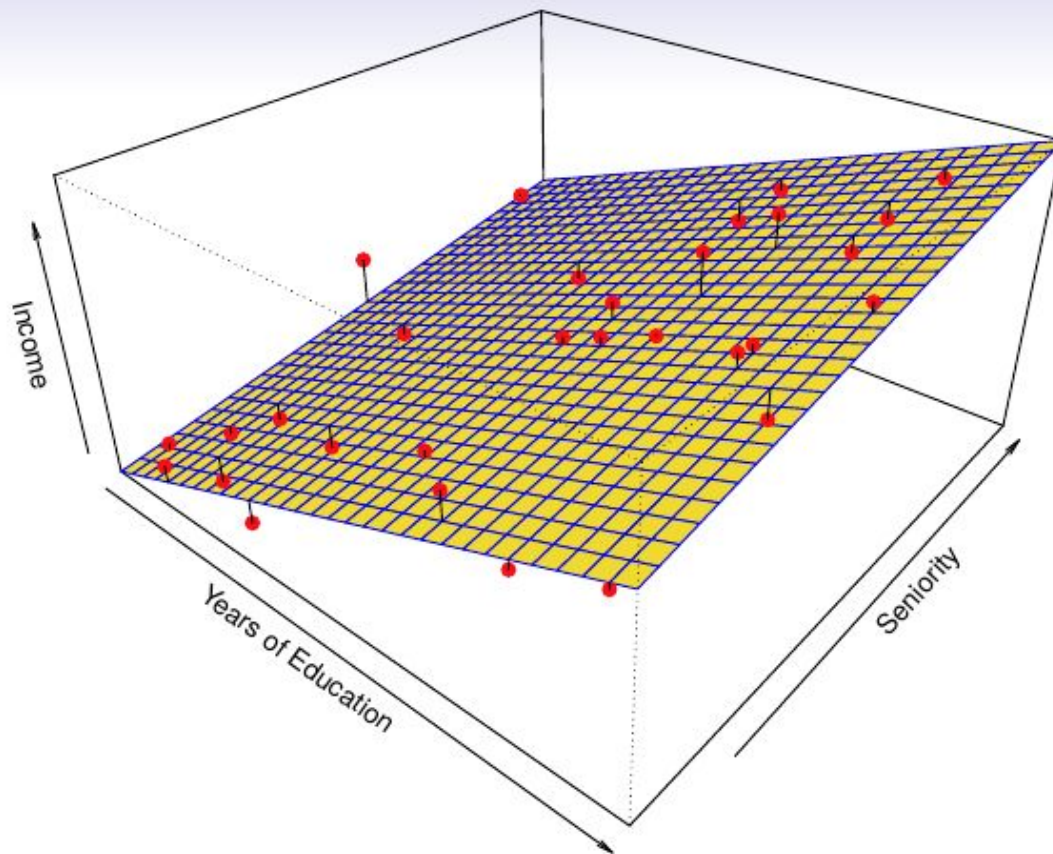




Simulated example. Red points are simulated values for **income** from the model

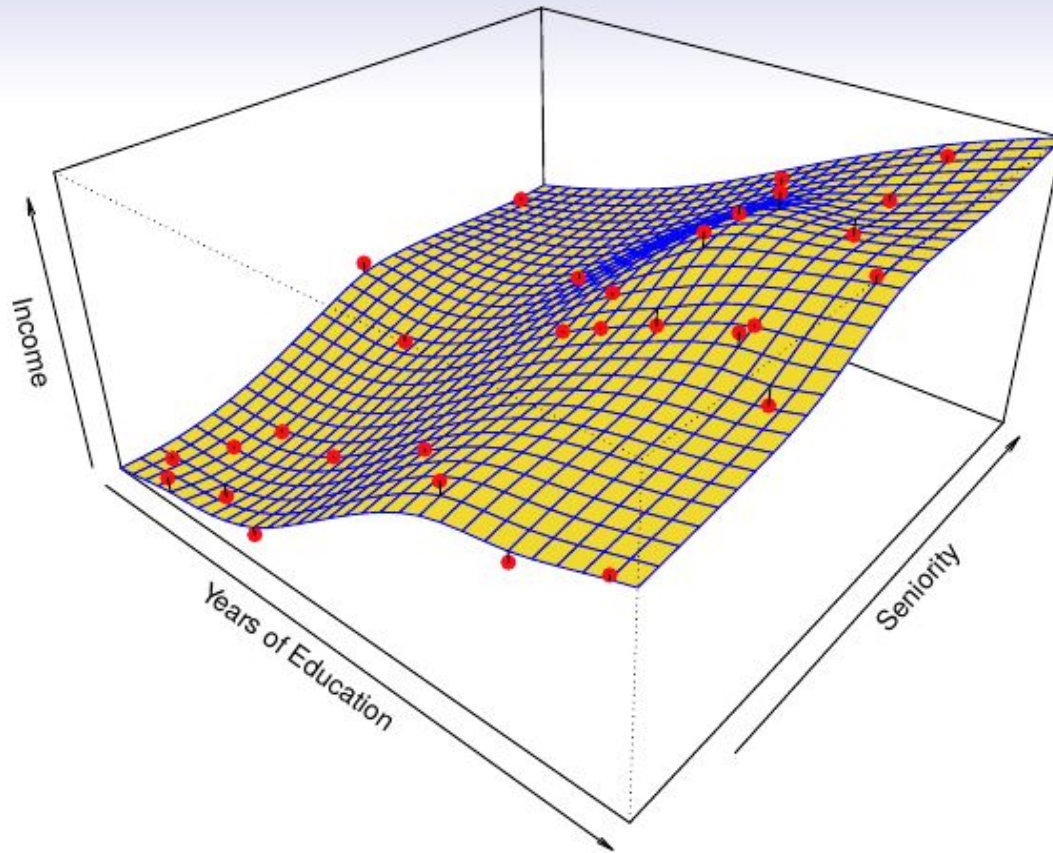
$$\text{income} = f(\text{education}, \text{seniority}) + \epsilon$$

f is the blue surface.

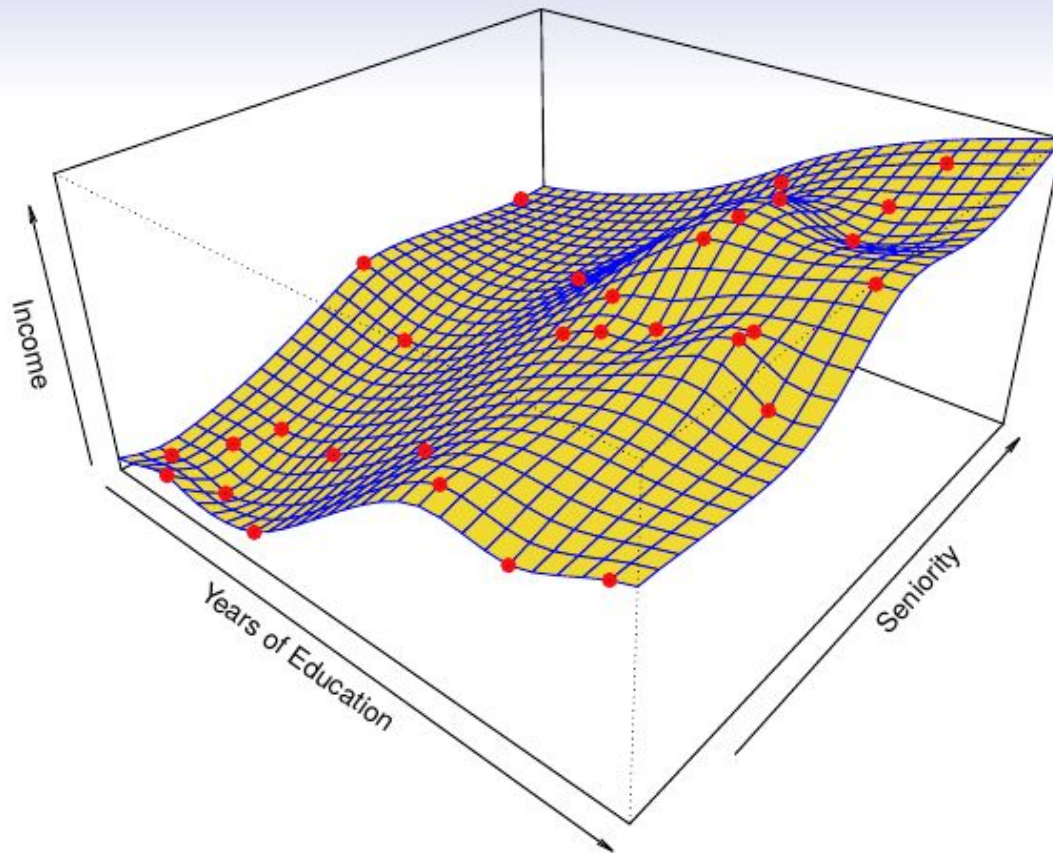


Linear regression model fit to the simulated data.

$$\hat{f}_L(\text{education}, \text{seniority}) = \hat{\beta}_0 + \hat{\beta}_1 \times \text{education} + \hat{\beta}_2 \times \text{seniority}$$



More flexible regression model $\hat{f}_S(\text{education}, \text{seniority})$ fit to the simulated data. Here we use a technique called a *thin-plate spline* to fit a flexible surface. We control the roughness of the fit (chapter 7).



Even more flexible spline regression model

$\hat{f}_S(\text{education}, \text{seniority})$ fit to the simulated data. Here the fitted model makes no errors on the training data! Also known as *overfitting*.

Some trade-offs

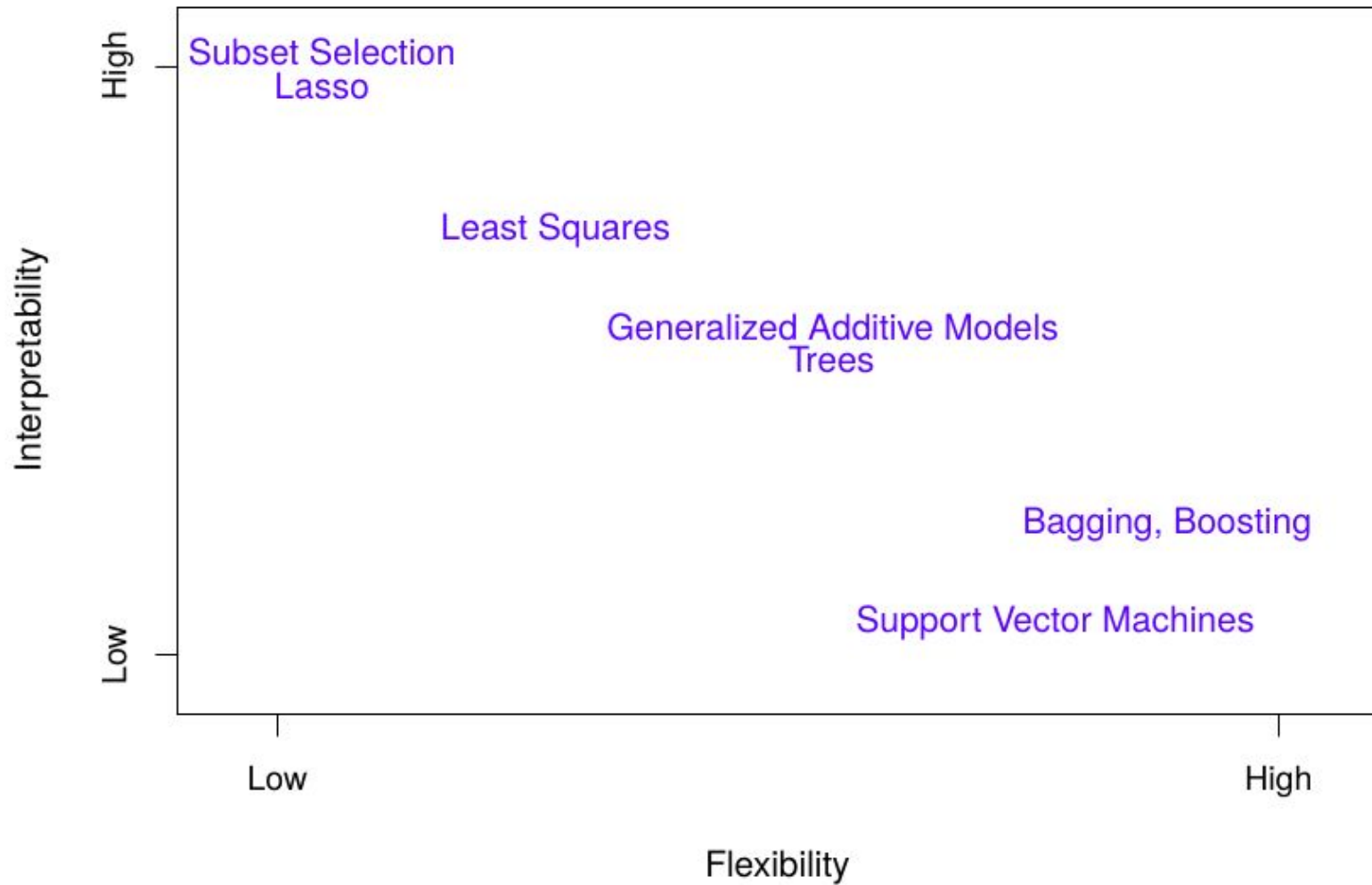
- Prediction accuracy versus interpretability.
 - Linear models are easy to interpret; thin-plate splines are not.

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- Prediction accuracy versus interpretability.
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- Good fit versus over-fit or under-fit.
 - How do we know when the fit is just right?

Some trade-offs

- Prediction accuracy versus interpretability.
 - Linear models are easy to interpret; thin-plate splines are not.
- Good fit versus over-fit or under-fit.
 - How do we know when the fit is just right?
- Parsimony versus black-box.
 - We often prefer a simpler model involving fewer variables over a black-box predictor involving them all.



Assessing Model Accuracy

Suppose we fit a model $\hat{f}(x)$ to some training data $\text{Tr} = \{x_i, y_i\}_1^N$, and we wish to see how well it performs.

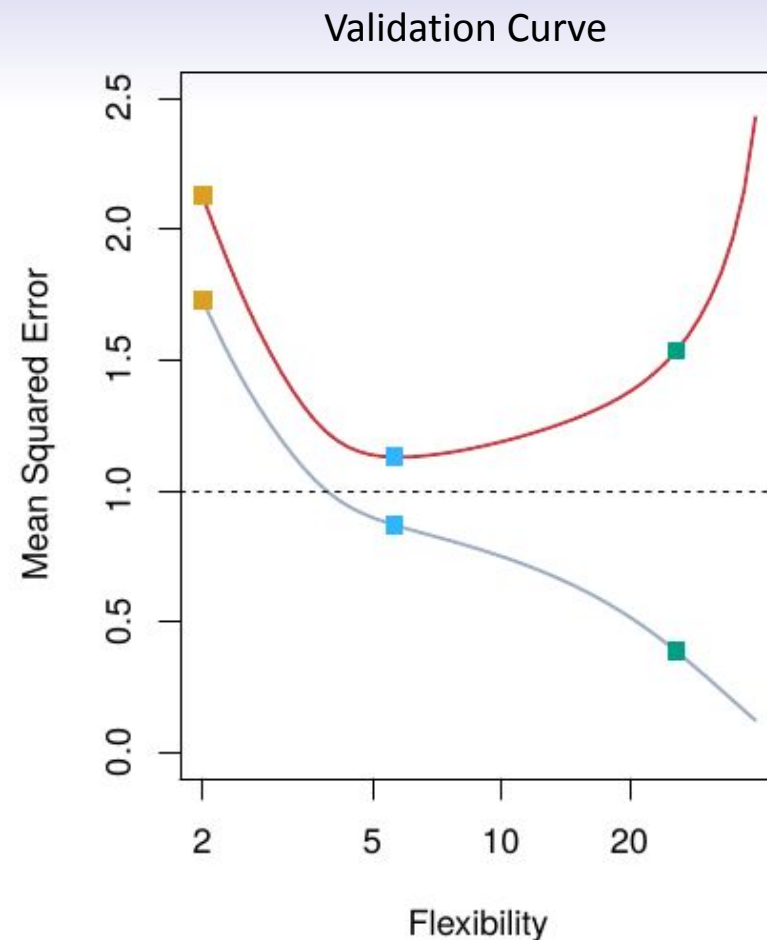
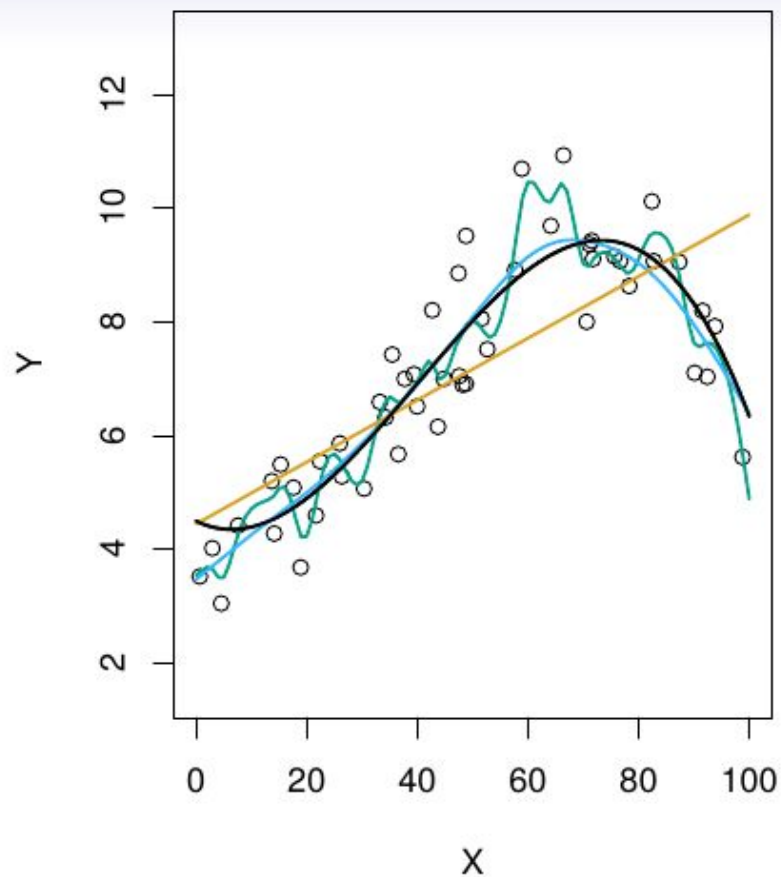
- We could compute the average squared prediction error over Tr :

$$\text{MSE}_{\text{Tr}} = \text{Ave}_{i \in \text{Tr}} [y_i - \hat{f}(x_i)]^2$$

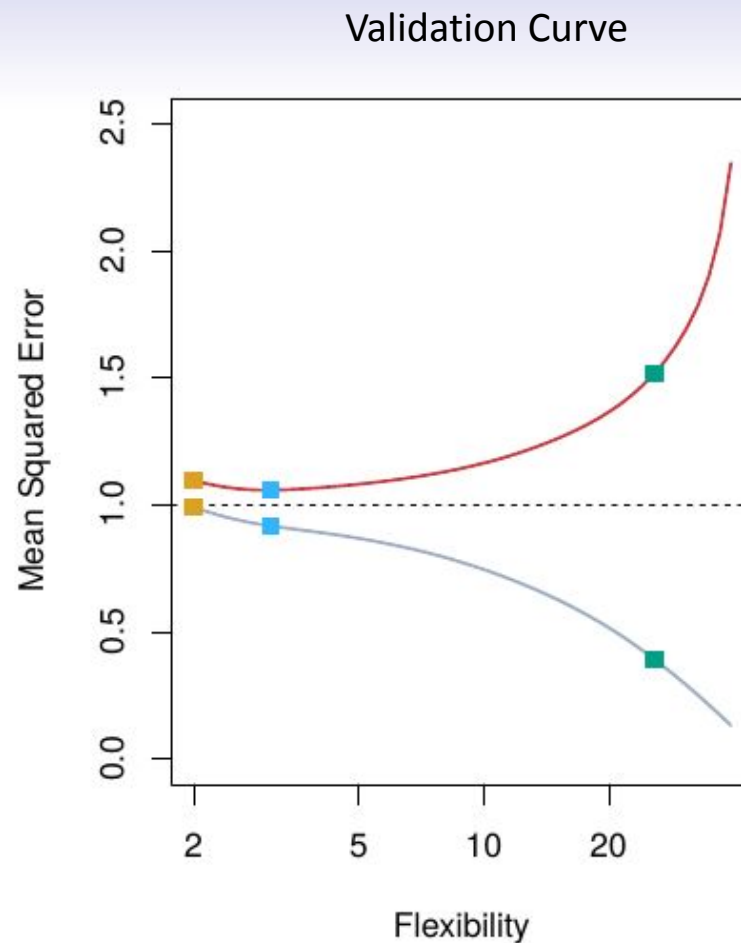
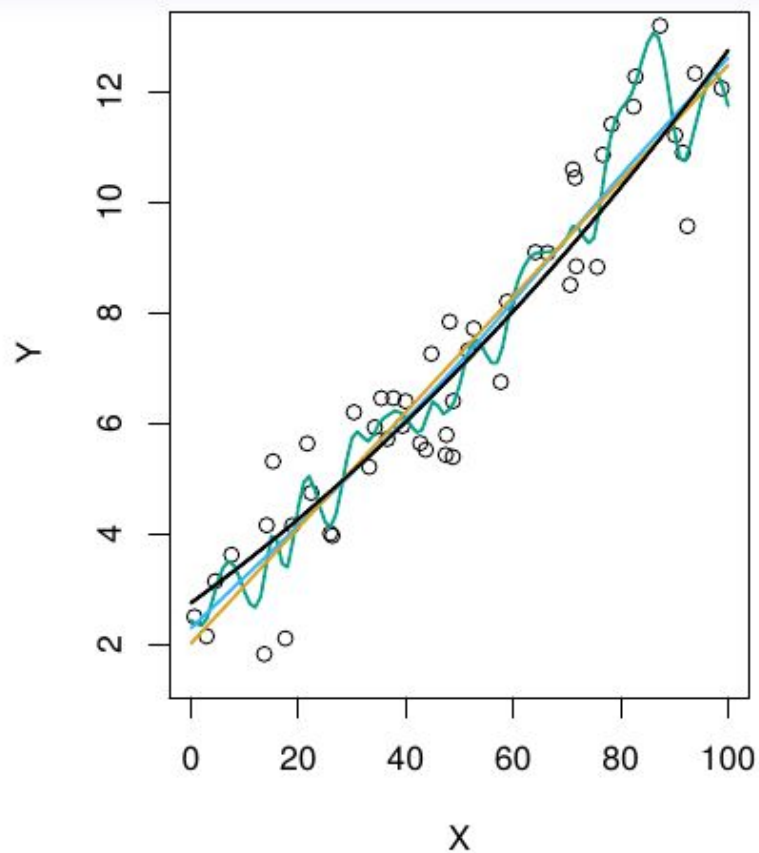
This may be biased toward more overfit models.

- Instead we should, if possible, compute it using fresh *test* data $\text{Te} = \{x_i, y_i\}_1^M$:

$$\text{MSE}_{\text{Te}} = \text{Ave}_{i \in \text{Te}} [y_i - \hat{f}(x_i)]^2$$

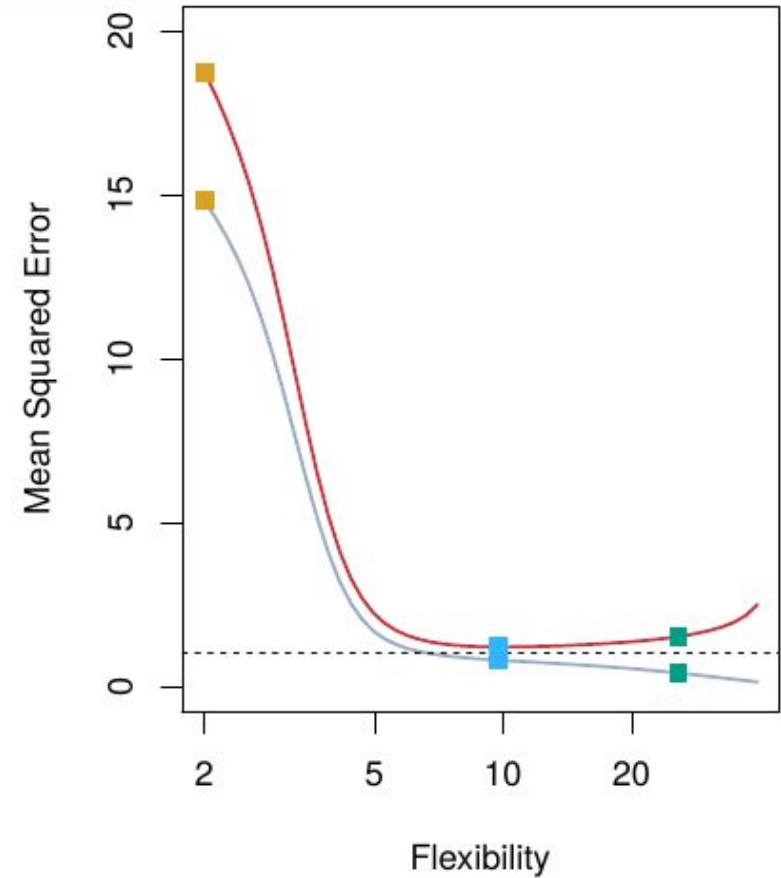
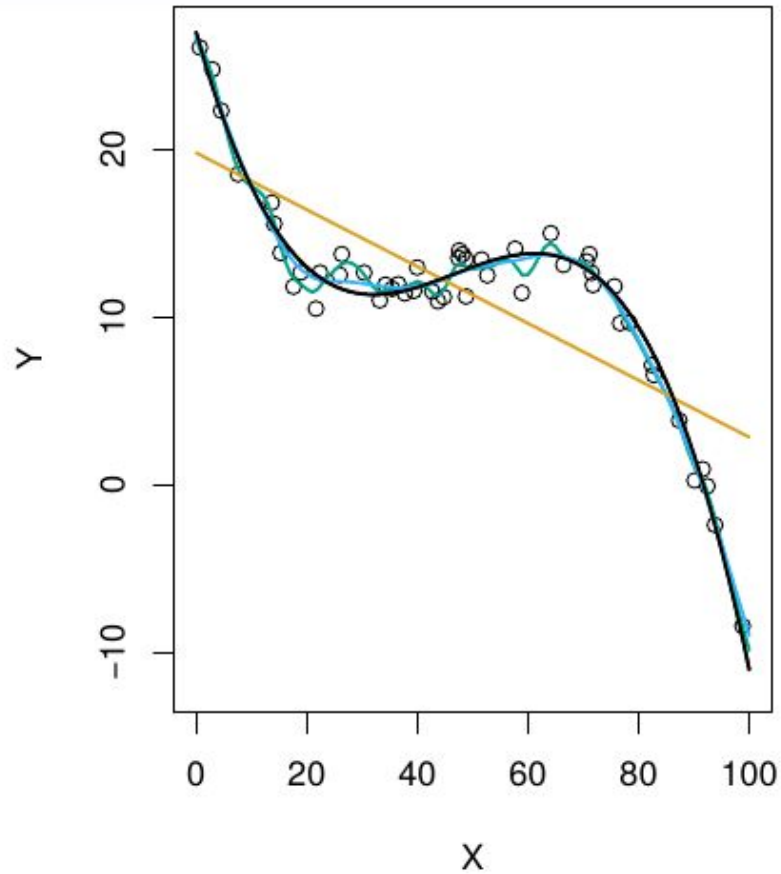


Black curve is truth. Red curve on right is MSE_{Te} , grey curve is MSE_{Tr} . Orange, blue and green curves/squares correspond to fits of different flexibility.



Here the truth is smoother, so the smoother fit and linear model do really well.

Validation Curve



Here the truth is wiggly and the noise is low, so the more flexible fits do the best.

Bias-Variance Trade-off

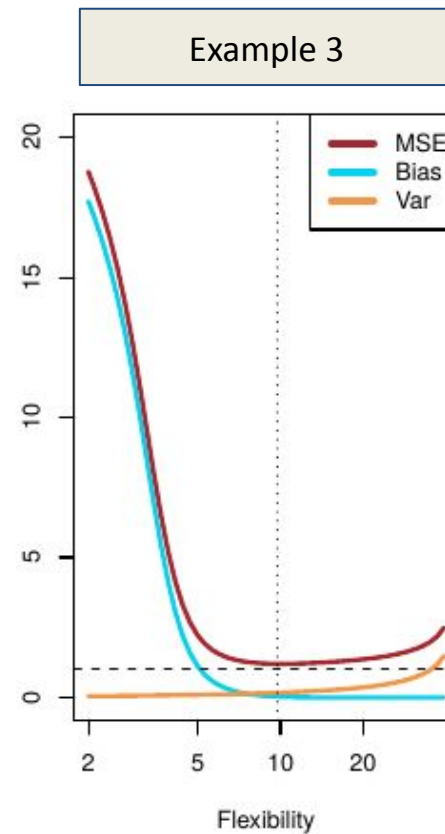
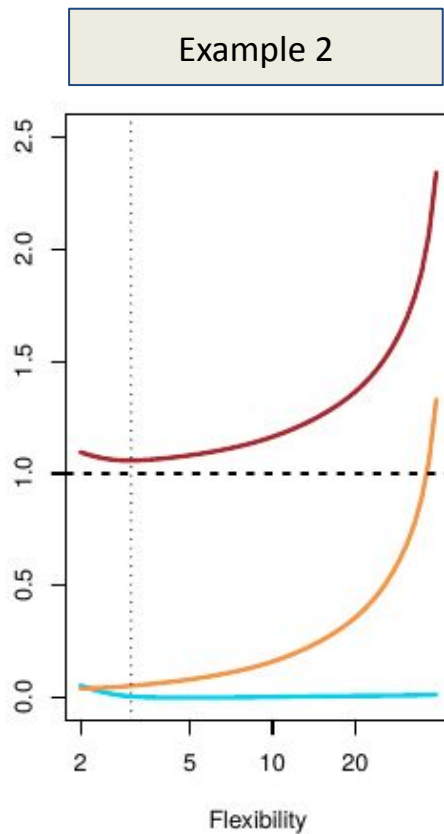
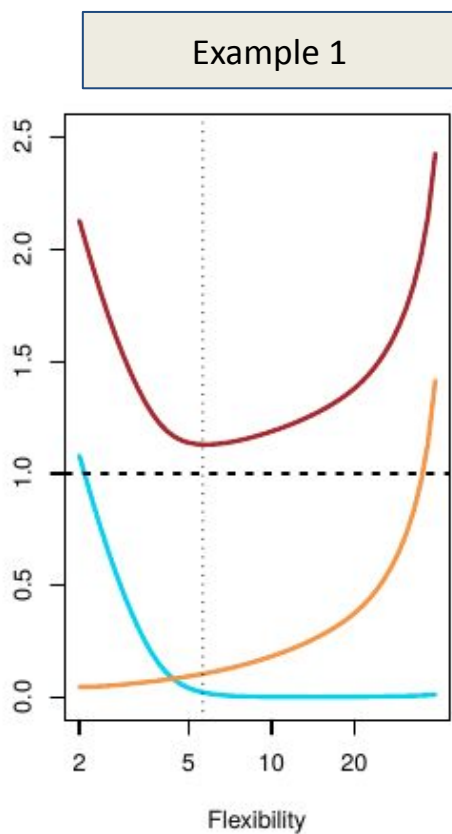
Suppose we have fit a model $\hat{f}(x)$ to some training data Tr , and let (x_0, y_0) be a test observation drawn from the population. If the true model is $Y = f(X) + \epsilon$ (with $f(x) = E(Y|X = x)$), then

$$E \left(y_0 - \hat{f}(x_0) \right)^2 = \text{Var}(\hat{f}(x_0)) + [\text{Bias}(\hat{f}(x_0))]^2 + \text{Var}(\epsilon).$$

The expectation averages over the variability of y_0 as well as the variability in Tr . Note that $\text{Bias}(\hat{f}(x_0)) = E[\hat{f}(x_0)] - f(x_0)$.

Typically as the *flexibility* of $\hat{f}$ increases, its variance increases, and its bias decreases. So choosing the flexibility based on average test error amounts to a *bias-variance trade-off*.

Bias-variance trade-off for the three examples

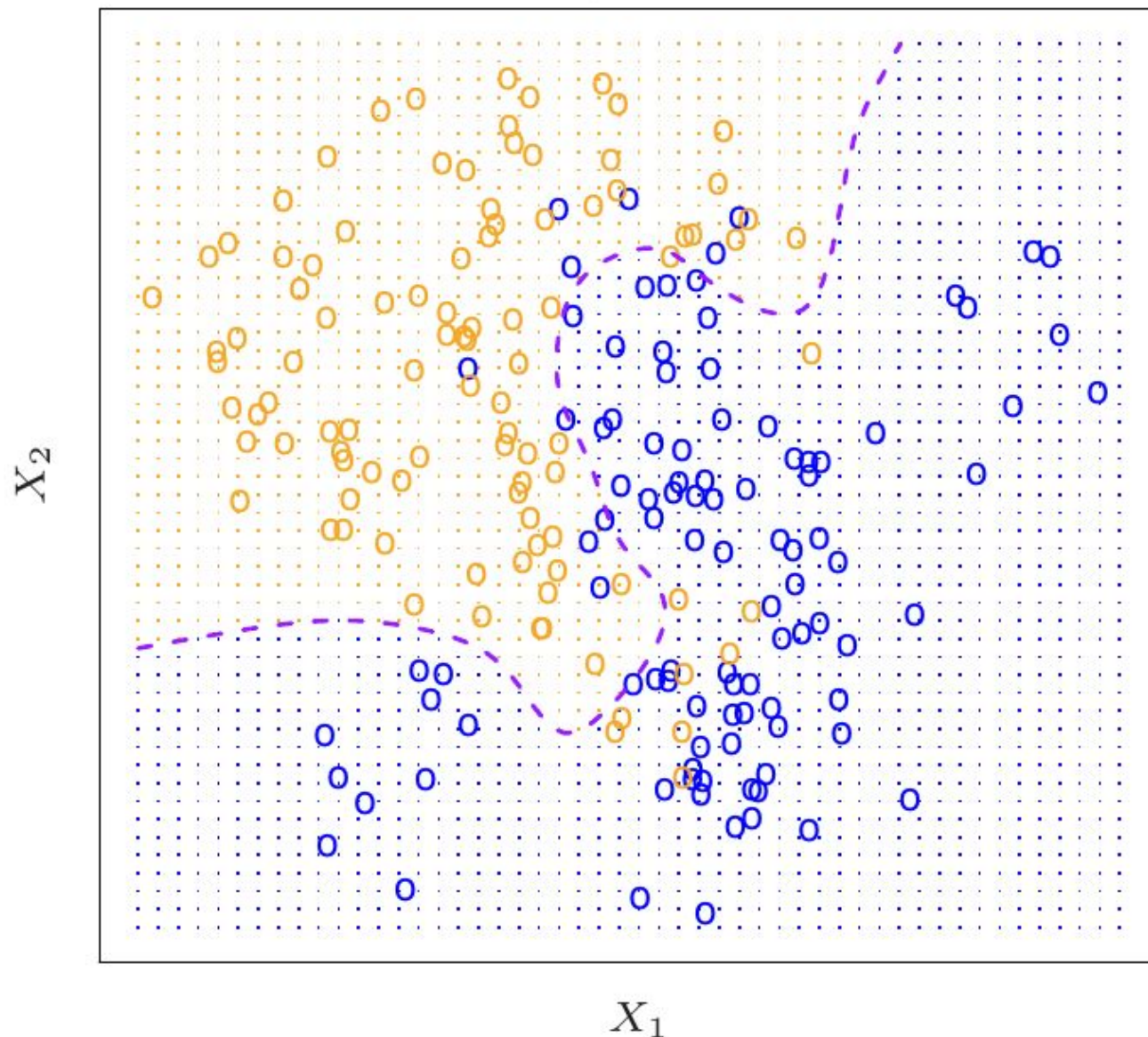


Performance of Classification Models

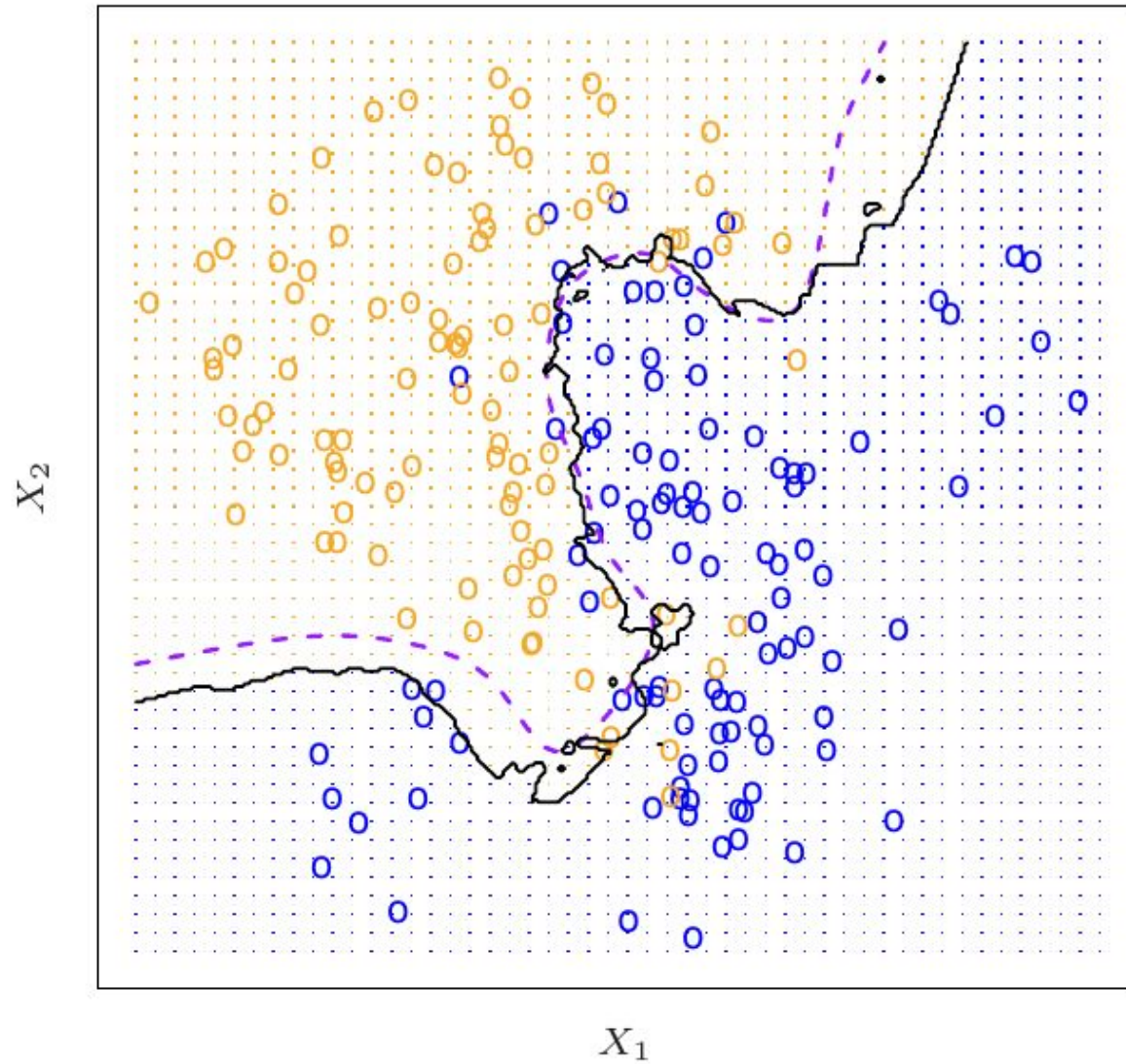
$$\text{Error Rate} = \frac{1}{n} \sum_{i=1}^n I(y_i \neq \hat{y}_i).$$

$I(y_i \neq \hat{y}_i)$ is an *indicator variable* that equals 1 if $y_i \neq \hat{y}_i$ and zero if $y_i = \hat{y}_i$.

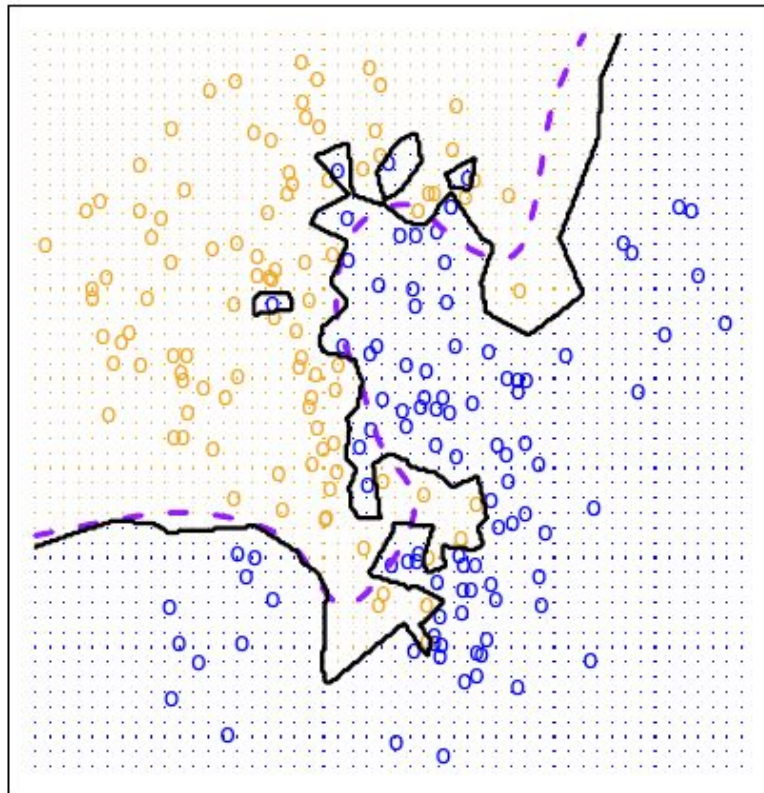
Example: K-nearest neighbors in two dimensions



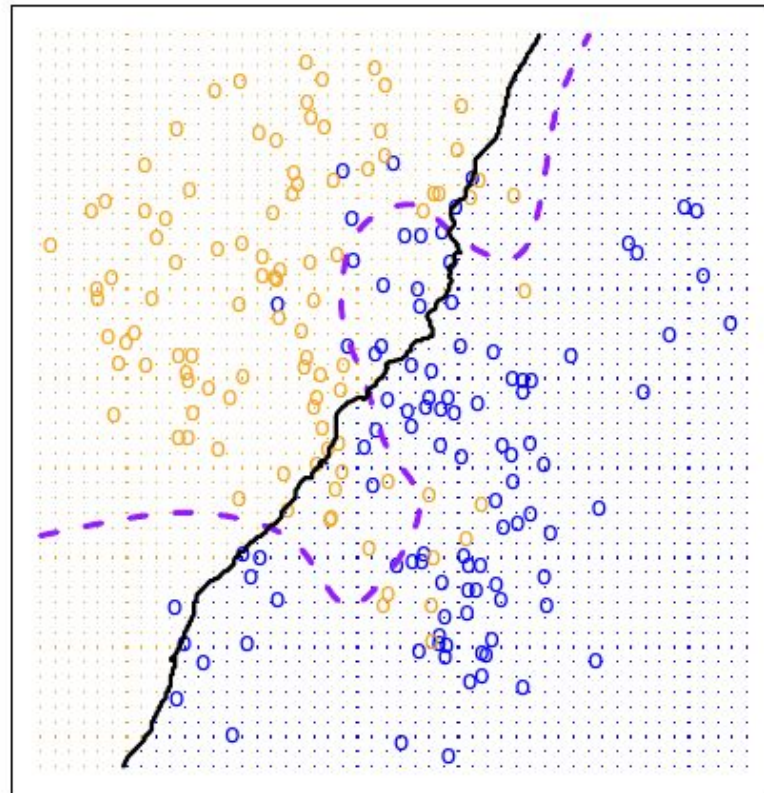
KNN: K=10

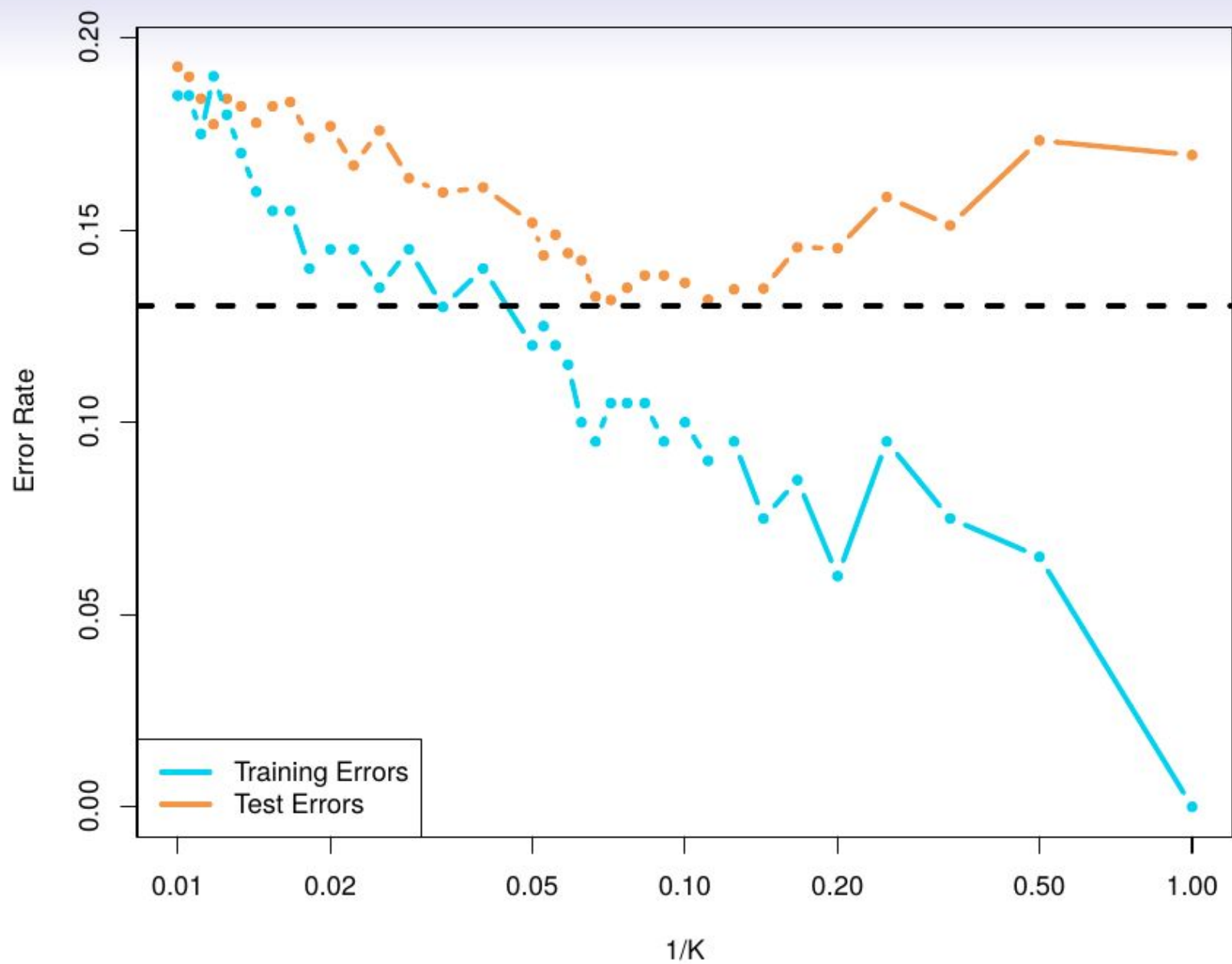


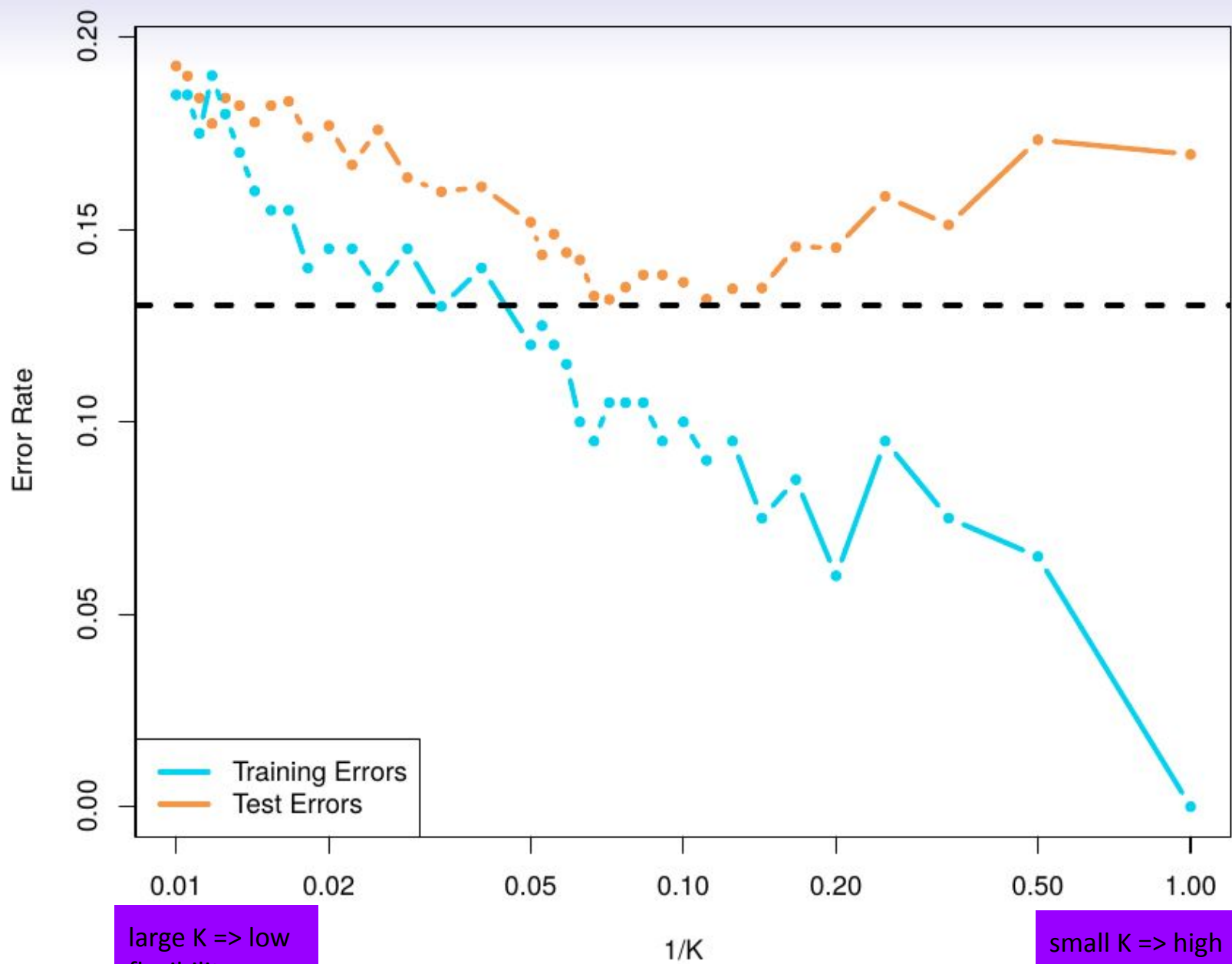
KNN: $K=1$



KNN: $K=100$







large K => low
flexibility

small K => high
flexibility